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Some Insights on Flow over Sharp-Crested Weirs Using Computational Fluid Dynamics: Implications for Enhanced Flow Measurement

Joseph M. Sinclair[1]; S. Karan Venayagamoorthy, Ph.D., A.M.ASCE[2]; and Timothy K. Gates, Ph.D., M.ASCE[3]

Abstract: This study reexamines flow over a sharp-crested weir using computational fluid dynamics (CFD) to identify the optimal operating range under which the weir functions with accuracy as a free-flowing measurement device. A numerical parametric study was conducted for two separate channel and weir geometries under a range of flow rates resulting in different h/P values, where h is the elevation head over the weir crest, and P is the weir height. Analysis of velocity and pressure profiles over the weir revealed three distinct flow regimes: a high acceleration regime, an ideal operating regime, and a weir-inundated regime that may lead to submerged flow. The high acceleration regime correlates to flows with $h/P < 0.6$, and the weir-inundated regime occurred when $h/P > 2.0$. The ideal operating regime was found to occur in the range $0.6 < h/P < 2.0$, where flow rate estimates using measurements of h are enhanced. These results are also consistent when considering how the discharge coefficient for the weir rating equation changes with h/P and highlight the need for caution in using empirical discharge coefficients over a broad range of h/P values. **DOI: 10.1061/(ASCE)IR.1943-4774.0001652.** *© 2022 American Society of Civil Engineers.*

Introduction

The sharp-crested weir is one of the simplest and most widely used hydraulic structures for flow measurement and regulation for many practical applications, such as in irrigation canals, ditches, and streams. It can be constructed in a wide variety of channels and is often used to control a number of different flow conditions. The sharp-crested weir's relatively high accuracy and ease of operation allow for its use to monitor flow rate and regulate upstream flow depth to best suit the desired conditions. Thus, it is not surprising that extensive research has been done on the sharp-crested weir ranging from experimental to field work to more recent computational fluid dynamics (CFD) studies.

Some of the earliest, best-known research dates back to the seminal work of Theodore Rehbock at Karlsruhe Laboratory in Germany wherein he developed an equation for estimating the discharge over the weir using nondimensional characteristics of the flow (King et al. 1929; Henderson 1966). Kindsvater and Carter (1957) proposed that viscous effects and surface tension effects can be accounted for in the discharge equation for a sharp-crested weir through minor corrections to the upstream head. A comprehensive account on weirs can be found in Ackers et al. (1978) and Bos (1989). Rajaratnam and Muralidhar (1971) conducted one of the earliest experimental studies to document the distribution of pressure and velocity over sharp-crested weirs. Their study highlighted the rather dramatic variations of both the pressure and velocity distributions over the crest of the weir and at the point of minimum energy head in the nappe (sheet of falling liquid) as a function of the nondimensional head over the weir crest (see also related work by Ramamurthy et al. 1987). More recently, Johnson (2000) experimentally analyzed the discharge coefficient for the sharp-crested weir equation for a variety of different conditions and identified the importance of including the upstream velocity head in computing the discharge coefficient. Other related experimental studies include those by Aydin et al. (2011) and Bagheri and Heidapour (2010), where laboratory experiments were used to study flow over rectangular sharp-crested weirs. In the last two decades, CFD has continued to advance and become an important tool in the analysis of hydraulic structures. Qu et al. (2009) were among the first to study flow over a sharp-crested weir using CFD. Abd El-Hady Rady (2011) used two-dimensional (2D) and three-dimensional (3D) models of sharp-crested weirs to highlight the importance of including the upstream velocity head in the discharge equation. Even more advanced (meshless) CFD techniques such as smoothed particle hydrodynamics have been used by Ferrari (2010) to study flow over sharp-crested weirs and by Mahdavi and Shahkarami (2020) to study flow over pivot weirs, which permit adjustment of the weir crest elevation. Other relevant CFD-based research of flow over weirs includes the recent studies by Altan-Sakarya et al. (2020), Aydin et al. (2016), Bilhan et al. (2018), Mahtabi and Arvanaghi (2018), and Saadatnejadgharahassanlou et al. (2020).

However, given the complexity of the actual flow over the sharp-crested weir and the underlying simplifying assumptions evoked for flow analysis, there remains quite a bit of uncertainty on the range of applicability of some of the well-known and commonly employed empirical formulations (e.g., King et al. 1929) for the discharge coefficient for sharp-crested weirs. As such, the purpose of this study was to reexamine flow over a sharp-crested weir

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using CFD to identify the optimal operating range under which the simplified weir discharge equation may be used with reasonable accuracy for flow measurement. Specifically, an overarching objective was to investigate the distribution of the nondimensional pressure and velocity profiles over the crest of the weir to provide some insights on the different flow states that are possible, their bearing on the relationship between flow rate and energy head, and the related implications for the calculation of the discharge coefficient. Additionally, in the spirit of this special issue, a secondary objective of this study was to identify best practices in setting up and running CFD open-channel flow simulations for irrigation and hydraulic engineering applications.

Theoretical and Numerical Considerations

Sharp-Crested Weir Theory

A sharp-crested weir is classically defined as a vertical plate perpendicular to the direction of flow with a crest width (thickness) (W_c) that is less than 2 mm in the flow direction. A broader definition of a sharp-crested weir is based on the ratio of the upstream total hydraulic head to the crest width $H/W_c > 1.5$ (see Hager and Schwalt 1994; Hager 2010), where $H = h + \alpha\bar{u}^2/2g$, $\bar{u}$ is the average of the velocity u oriented normal to the channel cross-section upstream of the weir, and α is the kinetic energy correction factor. Although H/W_c varies with flow conditions, the minimum value was 23 for all the cases in this study. Fig. 1 is a schematic that depicts the important study variables for a free-flowing sharp-crested weir. It is clear that the flow over a sharp-crested weir is complex given the pattern of the flow and the effects of turbulence and viscous processes that are extremely difficult to account for in simplified analysis. Regardless, taking the flow approaching and passing over the weir to be essentially one-dimensional, it is obvious that the discharge Q over the weir is driven by the elevation head h over the weir (or more precisely by the total head H). However, a simple relationship between these two variables is not readily possible given the aforementioned complexities. Thus, some rather grossly simplifying assumptions, namely (1) the kinetic energy head associated with the average approach velocity ($\bar{u}$) upstream of the weir is negligible; (2) atmospheric pressure exists on the nappe, both above and below; (3) the curvilinear nature of the flow over the crest is ignored, and thus the flow path is assumed to be horizontal over the crest; and (4) the effects of friction, turbulence, surface tension, and separated flow just upstream of the weir are negligible, are made to derive the following well known rectangular sharp-crested weir discharge equation (Bos 1989; Gerhart et al. 2021):

$$Q = \frac{2}{3} C_d b \sqrt{2g} h^{3/2} \tag{1}$$

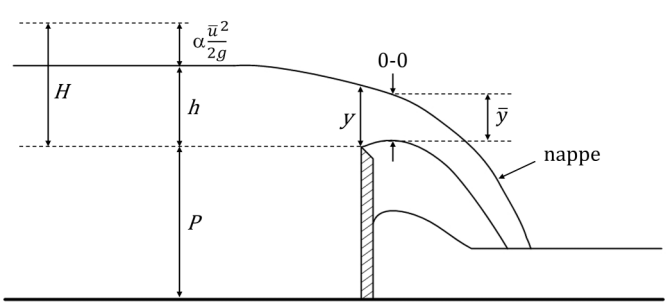

Fig. 1. Schematic of flow over a sharp-crested weir.

where C_d = discharge coefficient that has to be determined experimentally or through observation; g = acceleration due to gravity; b = length of the weir crest in the spanwise direction; and h = the elevation head, typically measured at 4 to 5 times the maximum expected h value upstream of the weir in order to avoid the acceleration zone near the weir. In Fig. 1, y is the depth directly above the crest, whereas $\bar{y}$ is the thickness of the nappe at Section 0-0. Section 0-0 is the point where the bottom of the nappe curve is at its highest elevation. The weir height P is measured from the base of the channel.

Eq. (1) follows directly from the Bernoulli equation under the simplifying assumptions noted previously. In order to account for these assumptions, an experimentally determined lumped parameter typically known as the discharge coefficient must be introduced. The discharge coefficient [as given in Eq. (1)] also accounts for the missing approach velocity head, which might not be negligible depending on the particular flow condition and the degree of crest contraction. A popular empirical formula for C_d is that due to King et al. (1929), where the effects of surface tension and viscous effects have been ignored and where the crest is fully suppressed (no lateral contraction from the channel sides). It has been suggested that this formula works well only for cases where pressure is atmospheric under the nappe and for h/P less than 2 to 5 (Bos 1989; Henderson 1966)

$$C_{dR} = 0.611 + 0.075\frac{h}{P} \tag{2}$$

Alternatively, some researchers (e.g., Johnson 2000) have analyzed the relationship between the discharge coefficient as a function of H/P rather than h/P. In this case, the weir discharge equation given in Eq. (1) would be revised as follows (Tracy 1957):

$$Q = \frac{2}{3} C_{dT} b \sqrt{2g} H^{3/2} \tag{3}$$

where C_{dT} = form of the discharge coefficient that captures the lumped effects of the simplifying assumptions stated previously but excluding the neglect of the upstream velocity head because it is explicitly accounted for in Eq. (3). However, it is oftentimes difficult to measure the velocity head, and hence this approach is not necessarily useful from a practical standpoint (Bos 1989). In this study, each of the three different discharge coefficients was analyzed to determine its dependence on h/P (or H/P for C_{dT}).

Finally, a simple dimensional analysis would reveal that the discharge coefficient should be a function of the Reynolds number (Re, accounting for viscous effects), the Weber number (We, accounting for surface tension effects), and h/P. Of these, h/P is the dominant parameter given that surface tension effects and viscous effects are only likely to be important at very low heads.

Computational Fluid Dynamics

In this study, computational fluid dynamics was used to analyze weir flow. Specifically, the CFD software called FLOW-3D (Flow Science 2021) was used given its ability to create more accurate free-surface models. Like other CFD solvers, FLOW-3D numerically solves the Navier–Stokes equations (momentum together with the continuity equations) to approximate incompressible fluid flow. It does so by using the finite-volume method, where the solution region is broken into smaller volumes, called cells. In FLOW-3D, the grids created are completely orthogonal, meaning each cell is a rectangular or cubic prism. The solver then analyzes each cell using boundary conditions or information from neighboring cells to solve for different fluid and flow variables. Scalar quantities are

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solved at cell centers, whereas vector quantities are solved on the cell faces. Fluid simulations can be one-dimensional, two-dimensional, or three-dimensional depending on the level of detail required. FLOW-3D also uses a unique volume-of-fluid (VOF) method (Hirt and Nichols 1981) that provides an accurate way to track and locate the free surface using structured meshes. This is achieved via a scheme to locate the free surface, an algorithm to track the free surface as a sharp interface, and application of boundary conditions at the surface.

FLOW-3D is also unique in the way it implements geometry in CFD problems. The fractional area-volume obstacle representation (FAVOR, Hirt 1993) technique calculates the volumetric and area ratios of the cells in order to embed different geometric obstacles. The upwind-differencing method is applied to approximate the momentum advection. The pressure is solved implicitly using the generalized minimum residual algorithm (GMRES) due to its high accuracy, rapid convergence, and parallel efficiency. The two-equation RNG k–ε closure model was used herein to model the effects of turbulence. Further information of the specifics of FLOW-3D's solver and other processes can be found in the online manual (Flow Science 2021). This code has been extensively validated for various environmental hydraulic flow problems (e.g., An et al. 2012; Zhou et al. 2017; Zhou and Venayagamoorthy 2019, 2020).

CFD Model Setup

Each CFD model presented in this study is a 2D model with only one cell in the lateral (spanwise) direction. Models were built using inspiration from the well-known experimental studies of Rajaratnam and Muralidhar (1971) in order to have accurate empirical weir and flume dimensions. Table 1 outlines the major simulation variables for two model categories: A and B. All category A models had the same flume and weir dimensions, whereas category B models had a different set of geometric values. The flume for category A simulations was 20 m long by 0.3 m wide with a weir height P of 0.3 m and thickness of 0.005 m. For category B simulations, the flume was 10 m long and 0.45 m wide with a weir height P of 0.03 m and thickness of 0.0015 m. The weirs were placed at 15 and 8 m downstream of the inlet for the A and B models, respectively. The vertical height of the simulation was varied depending on the expected h/P value.

Table 1. Parametric study parameters for flow over a sharp-crested weir

Trial name	H/P	h/P	Q (m^3/s)	P (m)	C_d	C_{dR}	C_{dT}
A1	0.388	0.378	0.03	0.3	0.887	0.633	0.854
A2	0.441	0.429	0.035	0.3	0.857	0.638	0.821
A3	0.492	0.477	0.04	0.3	0.834	0.642	0.796
A4	0.600	0.579	0.05	0.3	0.779	0.650	0.738
A5	0.647	0.625	0.0525	0.3	0.730	0.654	0.694
A6	0.684	0.660	0.0623	0.3	0.798	0.657	0.757
A7	0.705	0.679	0.06	0.3	0.738	0.658	0.697
A8	0.780	0.747	0.07	0.3	0.744	0.664	0.698
A9	0.853	0.813	0.08	0.3	0.750	0.669	0.698
A10	0.925	0.878	0.09	0.3	0.752	0.675	0.695
A11	0.985	0.930	0.1	0.3	0.766	0.679	0.702
A12	1.059	0.997	0.11	0.3	0.759	0.685	0.693
A13	1.132	1.063	0.12	0.3	0.753	0.690	0.684
A14	1.382	1.280	0.16	0.3	0.759	0.708	0.677
A15	1.745	1.596	0.22	0.3	0.750	0.734	0.656
A16	2.160	1.944	0.3	0.3	0.761	0.763	0.649
A17	2.365	2.129	0.334	0.3	0.739	0.779	0.631
A18	2.681	2.375	0.41	0.3	0.770	0.799	0.641
B1	3.125	2.624	0.027	0.03	0.920	0.808	0.708
B2	3.770	3.129	0.035	0.03	0.916	0.846	0.693
B3	4.325	3.56	0.042	0.03	0.905	0.878	0.676
B4	4.939	4.041	0.05	0.03	0.892	0.914	0.660
B5	5.615	4.545	0.06	0.03	0.897	0.952	0.653

The actual experimental flumes employed by Rajaratnam and Muralidhar (1971) were only 5 m long. The increased upstream distance and extended flume length in the CFD models were used to allow for flow development so as to create as accurate a simulation as possible. Unlike laboratory flumes or some other CFD codes, it is not possible to specify a fully developed turbulent boundary layer flow at the upstream (inlet) boundary in FLOW-3D. Hence, in order to achieve fully developed conditions well upstream of the weir, sufficient distance upstream was provided. A verification trial was conducted to check if flow development was important, and it was found that exactly mimicking laboratory flume dimensions without accounting for flow development produced more inaccurate results.

For the category A simulations, the resolution of the mesh in the x (streamwise) direction was 10 mm, whereas the resolution in the wall-normal (y) direction was 5 mm. The category B simulations had a resolution of 5 mm in the streamwise (x) direction and a resolution of 2.5 mm in the wall-normal (y) direction. As mentioned earlier, the models are all 2D, so the spanwise (z) direction is composed of a single cell. A small sensitivity study was run to reduce and increase the mesh resolution. Increasing the resolution to a finer scale often led to time-step errors in the solver, which would cause the simulation to terminate. A coarser resolution led to less accurate results. Hence, the resolution was selected to be as fine as possible without causing time-step errors in order to gather more data if it were needed.

Two-dimensional, as opposed to 3D, simulations were chosen for two reasons: to reduce computational cost and to increase theoretical accuracy. A third dimension exponentially increases the computational time required to run a simulation and, given the number of simulations required for this parametric study, it was not feasible to run upwards of fifteen 3D simulations in the given time frame. Also, the intent of the flume studies in Rajaratnam and Muralidhar (1971), as well as the present study, was to be independent of the spanwise direction. That is, the lateral walls were designed to have minimal influence on the flow in the flume. In Rajaratnam and Muralidhar's (1971) studies, the flume was made of smooth glass in an attempt to reduce the frictional effects of the lateral walls. In CFD, a 2D model is the easiest method for eliminating lateral effects.

In terms of boundary conditions, at the upstream end, a volumetric inflow condition was used, where a steady flow rate was specified for the duration of the simulation. The downstream end was either a prescribed outflow or a pressure outlet condition with a specified fluid height. The outflow condition allows fluid to exit the simulation without any backflow and was used where the weir could act as a hydraulic control. However, when the weir could not act as a control or there was the possibility of submerged flow, the pressure outlet condition was used to control flow. The lateral sides of the flume were prescribed as symmetry boundary conditions. The symmetry condition does not allow lateral flux of fluid and does not create boundary layer flow like a wall. This greatly reduces the computational cost while also increasing the accuracy of the simulations. The bed of the flume was specified as a wall condition where the no-slip rule was enforced, but with zero roughness height and zero slope. Most flumes are smooth walled and configured to reduce the roughness height as much as possible, but with a CFD model, true zero roughness height can be specified. Finally, the topmost vertical condition was prescribed as a pressure fluid–fraction condition. The pressure fluid–fraction condition assumes that the fluid fraction at the given boundary is zero; that is, no fluid is present. However, this upper boundary condition is not

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relevant because the height of the simulated flume is such that the fluid will never reach the upper boundary. As a result, the free surface can develop without additional influences. The upper boundary condition is merely a failsafe to check in case something is wrong with the simulation.

In FLOW-3D, the velocity in the first cell adjacent to a wall is approximated using the log-law. The first cell height should fully encompass the viscous region and end within the log region. For that reason, the nondimensional distance from the wall measured in viscous lengths (or wall units) y^+ of the first cell is recommended to be greater than 30 but less than 100 to 500, depending on the flow. Multiple simulations were conducted for the flume configurations mentioned previously to identify what cell resolution would work best for the flow rate and geometry used. It was found that the first cell height needed to be greater than 2 but less than 10 mm on average.

To better resolve the weir geometry, a nested mesh was used. A nested mesh in FLOW-3D is a smaller mesh block that typically has finer resolution and sits within a larger mesh block. It can be used to better resolve geometry (such as that of a sharp-crested weir) or to provide more accurate information in flow regions that are of higher importance. Here, the nested mesh was used to provide more cell faces and cell junctions so that FAVOR could adequately resolve the weir geometry. The nested mesh in the parametric study was the same thickness, height, and width of the weir and split the vertical and horizontal cells of the outer flume mesh in half, doubling the resolution. This ensured that no fluid flow occurred within the geometry. The nested mesh's sole purpose was to better resolve the weir geometry.

Verification and Validation

As best practice, it is important to verify numerical models for compliance with fluid laws and known theoretical results. One example is a conservation of mass check. With the models set as they are, there is a fixed amount of volumetric flow that can occur at any location along the lengthwise flume direction x. Calculations were made using the flow velocity and depth at certain x-coordinates to identify if the fixed flow rate was maintained; that is, no mass had entered or exited the simulation through any holes in the geometry or numerical errors in the model. Furthermore, it was important to verify that each simulation had a fully developed flow upstream of the weir. The weir discharge equations function under the assumption that flow is well developed, hence the extended upstream distance used in the CFD models. Fig. 2 shows the velocity and pressure profiles at three upstream locations for $h/P = 0.38$. It can be seen that the flow reached fully developed conditions upstream of the weir and that the pressure distributions were hydrostatic, as expected. A similar procedure was followed to ensure fully developed flow upstream of the weir for all simulations.

Table 2. Experimental parameters from Rajaratnam and Muralidhar (1971) used for validation

Experiment name	P (m)	h (m)	h/P	Q (m^3/s)
$A1^E$	0.3	0.185	0.625	0.0525
$A2^E$	0.3	0.207	0.697	0.0623
$B1^E$	0.175	0.198	1.122	0.0606
$C3^E$	0.03	0.099	3.440	0.0420

In addition to verification, validation with well-established experimental or field results is another best practice in CFD. Validation was done using the experimental study results of Rajaratnam and Muralidhar (1971). Nondimensional analysis for the pressure and velocity distributions above the weir crest and at Section 0-0 for a number of different h/P values was presented in their experimental study. Table 2 summarizes the key experimental parameters that were used in creating CFD models of the experimental trials. Four experiments in total were used for validation given the range of h/P values they capture. It should be noted that experiment $C3^E$ shown in Table 1 was conducted using a slightly longer and wider flume than the flume that was used for cases A and B in the study by Rajaratnam and Muralidhar (1971).

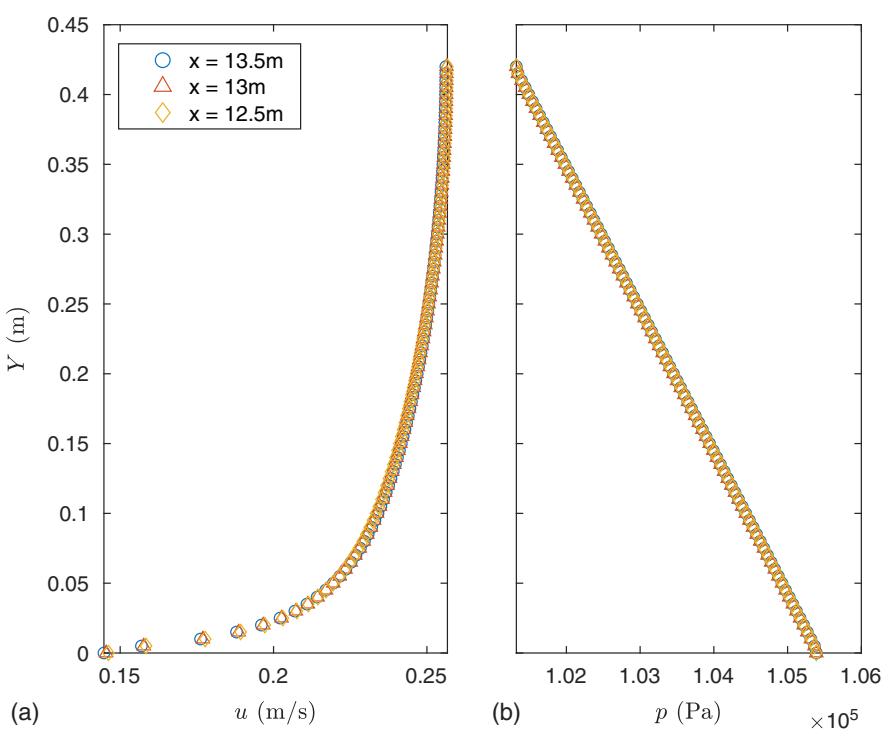

Fig. 2. (Color) (a) Velocity profiles; and (b) pressure profiles at three different locations upstream of the weir for $h/P = 0.38$.

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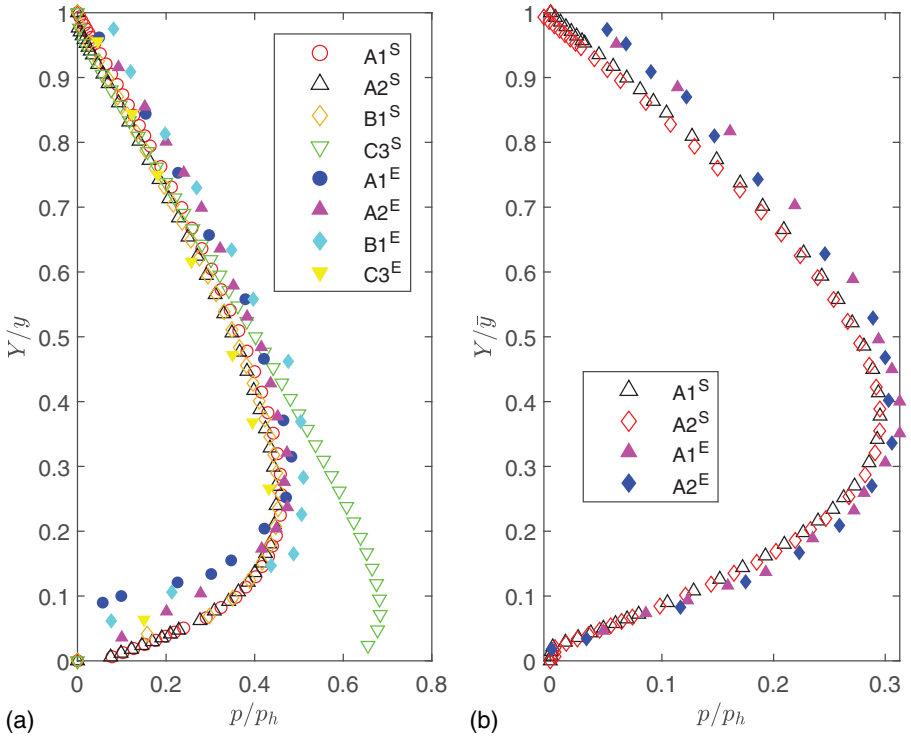

Fig. 3. (Color) Nondimensional pressure profiles at (a) weir crest; and (b) Section 0-0 comparing CFD simulations (denoted using the "S" superscript in the legends) with experimental results.

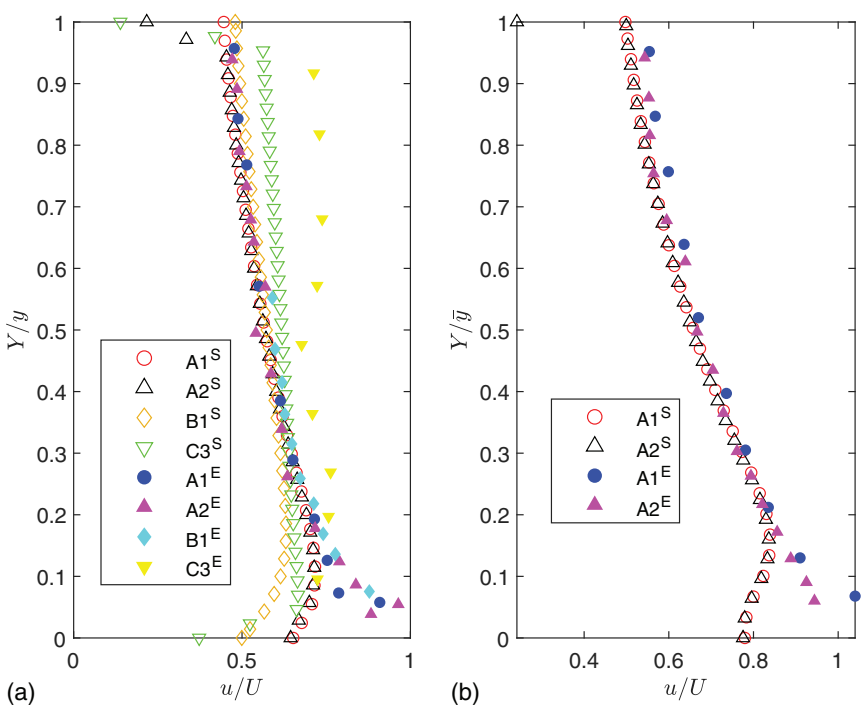

Fig. 4. (Color) Nondimensional velocity profiles at (a) weir crest; and (b) Section 0-0 comparing CFD simulations (denoted using the "S" superscript in the legends) with experimental results.

In addition to the values provided in Table 2, nondimensional velocity and pressure values were digitized from graphs presented in the study by Rajaratnam and Muralidhar (1971). The experimental models were simulated in CFD and results were compared to measured pressure and velocity distributions, as shown in Figs. 3 and 4, respectively. Pressure was nondimensionalized using the hydrostatic pressure p_h at either the crest or Section 0-0 location. At the crest, this value is equal to γy, and at Section 0-0, the value is $\gamma \bar{y}$, where γ is the specific weight of water. Lengthwise velocity was made nondimensional using a reference

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(theoretical) velocity, denoted as U, equal to $\sqrt{2gh}$, where h is the upstream elevation head above the weir crest. Vertical Y values, indicating vertical position within the nappe measured from the crest or bottom of the nappe, were nondimensionalized using either y or $\bar{y}$, depending on whether analysis was at the crest or Section 0-0, respectively.

The distribution of the pressure followed a similar shape both at the weir crest and at Section 0-0, as found in the experiments, starting at roughly 0 gauge pressure at the weir crest or nappe bottom and increasing until a maximum at a Y/y (or $Y/\bar{y}$) value between 0.2 and 0.3 before decreasing to 0 gauge pressure where the fluid met the free surface. The differences between the CFD and experimental results were most noticeable at the lower nondimensional y-values, especially for Case C3^{E}. This may be attributed to the inherent difficulty in experimentally measuring pressure in this active curvilinear zone of flow as well as challenges in accurately capturing this zone in CFD. However, the differences decreased significantly with an increase in nondimensional y-value, and the profiles compare favorably for the most part.

The shape of the velocity profiles also differs at the lower y-values. The experimental results revealed a continual increase in velocity at the lower Y/y (or $Y/\bar{y}$) values, whereas the CFD simulated velocity reached a maximum value before decreasing. In actuality, the horizontal velocity should be zero at the weir crest due to the no-slip condition. Analysis using potential flow theory (albeit for the wrong reason) also shows that the horizontal velocity over the crest should be zero because the flow is vertical on the face of the weir (McNown et al. 1955). Therefore, the CFD results appear more accurate shapewise based on such considerations. However, the overall agreement between the results is reasonable.

After analyzing these findings, it was concluded that the CFD models provided reasonably accurate representations of experimental hydraulic flows over sharp-crested weirs. The parametric study analyzing the different variables was for Q and h/P values within the validated range so as to have confidence in the results. Each simulation was also analyzed using the same verification methods as listed previously to ensure they were physically reasonable.

Results and Discussion

The main simulation data are given in Table 1. All three discharge coefficients are presented: C_d using Eq. (1), C_{dR} using Eq. (2), and C_{dT} using Eq. (3), respectively. Nondimensional velocity and pressure profiles are presented in Figs. 5 and 6, respectively, with the straight line in Fig. 6 representing the hydrostatic pressure distribution.

Analysis of both the velocity and pressure profiles over the crest of a sharp-crested weir revealed at least three distinct regimes. The first was flow with $h/P \leq 0.6$, which was named the high acceleration regime. Here, the velocity profiles (see Fig. 5) showed a strong convective acceleration, with higher relative velocities close to the weir crest. As a result of this, pressure profiles in this subregime experienced a negative relative pressure close to the weir crest. As h/P increased, velocity profiles displayed a decrease in relative magnitude, with the acceleration zone close to the weir crest shrinking. Similarly, pressure profiles, as shown in Fig. 6, close to the weir crest became less negative and approached atmospheric pressure. Flow could often be seen clinging to the weir in some of the lower $h/P(<0.5)$ simulations, indicative of the relatively strong influence of surface tension and viscous effects and mimicking free overfall conditions.

The second regime occurred for $0.6 < h/P \leq 2.0$ and was named the ideal operating regime. Here, both velocity and pressure

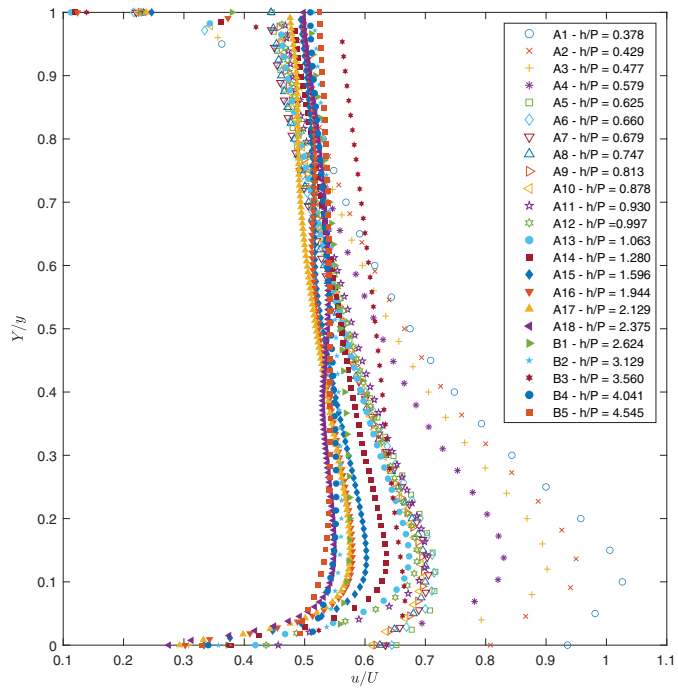

Fig. 5. (Color) Nondimensional velocity profiles over weir crest.

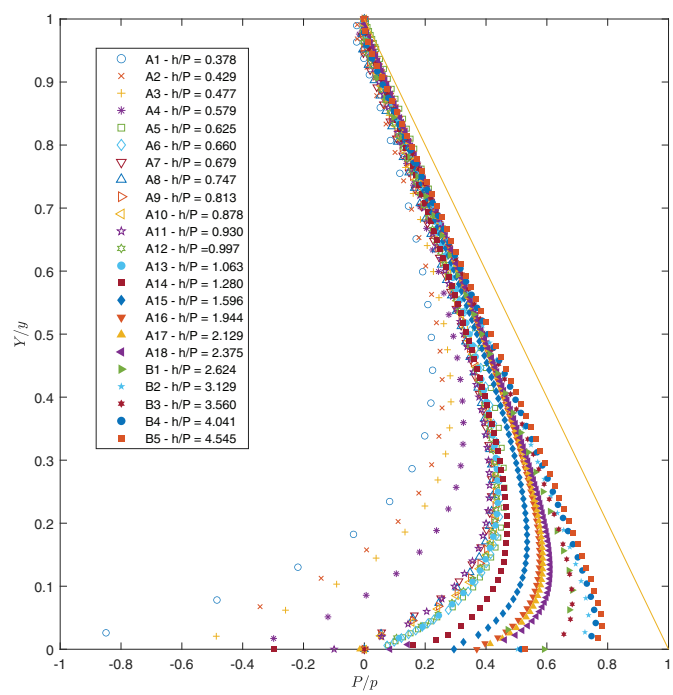

Fig. 6. (Color) Nondimensional pressure profiles over weir crest.

profiles collapsed on one another for the most part. Velocity profiles had a smaller relative acceleration zone close to the weir crest before decreasing linearly at higher nondimensional Y/y values. Pressure profiles started at atmospheric pressure at the weir crest before increasing until a maximum nondimensional pressure at a nondimensional Y/y value of about 0.15. The nondimensional pressure then decreased linearly, with increasing distance from the crest. It was in this regime that flow was well behaved (i.e., free-flow) according to the assumptions in the derivation of the discharge equation.

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Beyond a h/P value of about 2, both the velocity and pressure profiles began to diverge from the collapsed region. Velocity profiles continued to decrease in relative magnitude close to the weir crest, and the profiles began to look like a turbulent boundary layer flow with a developing region close to the weir crest and the velocity experiencing a smaller linear change at higher nondimensional Y/y values. The pressure profiles began to increase in magnitude closer to the weir crest and approached a more hydrostatic profile as h/P increased. In this region, the effects of the weir began to be drowned out as the structure was inundated by high flow. Beyond a h/P value of 3, the weir became submerged, with water levels rising above the weir crest on the downstream side to create a backwater effect. Thus, this regime was termed the weir-inundated regime.

Based on each of these regimes, the ideal operating range for a sharp-crested weir appears to fall in the range $0.6 < h/P \leq 2.0$. Here, high flow acceleration is not a significant factor in flow properties over the weir crest, and h/P is still not excessively large such that the weir continues to act with influence on the flow. Outside of this regime, additional irregularities are introduced in the form of negative gauge pressures and flow drowning out the weir's effect, making it unsuited for accurate flow regulation or measurement.

The three regimes can also be seen in plots of the calculated discharge coefficients, presented in Fig. 7. Both C_d and C_{dT} experienced a linear downward trend in the first regime for $h/P \leq 0.6$. The discharge coefficients then became relatively constant for the second regime ($0.6 < h/P \leq 2$) but with C_d at a higher level of about 0.75 compared to the C_{dT} value hovering just below 0.7. Here, the standard deviation was only about 0.02 for both coefficients. The C_{dT} trends for $H/P < 1.2$ matched well with experimental results from Johnson (2000). Beyond an h/P (or H/P) of 2.0, the discharge coefficients trended lower in magnitude before jumping up at higher values of h/P (or H/P). Beyond an h/P value of 3, it was observed that the flow was submerged and downstream conditions affected upstream flow. Calculation of the discharge coefficient was done in the same manner as previous calculations without the introduction of an empirical submergence factor. This was intentional in order to highlight how the discharge coefficients calculated using the simple weir flow given by either Eq. (1) or Eq. (3) differ considerably from typical values obtained for ideal weir flow conditions, due to the dramatically different submerged flow conditions. The purpose of this study was to look more theoretically at the sharp-crested weir without the influence of empirical effects. Therefore, selecting a submergence coefficient or a method for calculating the submergence coefficient would introduce too many extraneous factors.

The discharge coefficient C_{dR} (calculated from the Rehbock equation) only increased linearly with increasing h/P values. This shows a stark contrast to the other coefficients in trend and little coincidental agreement. Henderson (1966) and others noted the Rehbock equation can be used for h/P up to 5 with good agreement. However, these results indicate that approximating the discharge coefficient as a positive linear trend over this range results in inaccurate values for the majority of h/P values.

Furthermore, in the ideal operating regime, it appears that assuming a constant value for the discharge coefficient C_d might be helpful for practical purposes. From Fig. 7, it appears that using a value of $C_d = 0.75$ might be a good approximation. Fig. 8 shows excellent agreement between the predicted discharge using Eq. (1) with a discharge coefficient of 0.75 versus the actual discharge represented in the CFD model for the ideal operating regime ($0.6 < h/P < 2.0$). Also shown in Fig. 8 is the predicted discharge based on the coefficient of discharge calculated using the Rehbock equation [Eq. (2)]. The predicted discharge is lower than the actual discharge by up to 17% at lower h/P compared to about 6% when using a constant C_d of 0.75.

Visual changes in flow can also be observed from postprocessed colored contour plots of the simulation results. Figs. 9–11 show contour plots of the magnitude of the simulated resultant velocity of the flow region close to the weir. Fig. 9 is for $h/P = 0.38$, in the high acceleration regime. Here, the flow can be seen clinging to the weir due to the negative pressure underneath the nappe. The flow field for h/P of 0.81 is shown in Fig. 10 and is within the ideal operating regime. The nappe is well formed and aerated underneath, as assumed in the derivation of weir discharge equation. The inundated flow regime is depicted in Fig. 11 for h/P of 4.55 with submerged flow. In such a situation, the weir is effectively no longer useful as a flow measurement device because it is not able to act as an effective control, instead acting as a sill on the channel bed.

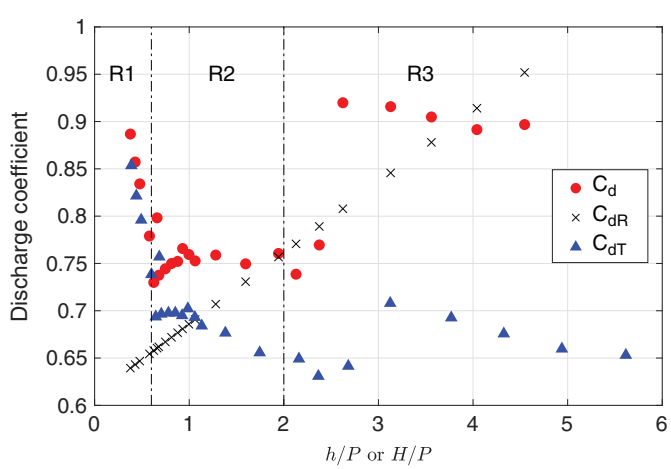

Fig. 7. (Color) Discharge coefficients as function of h/P (for C_d and C_{dR}) and H/P for C_{dT}. R1 denotes the high-acceleration flow regime, R2 is the ideal operating flow regime with a free flow over the weir, and R3 denotes the weir-inundated flow regime.

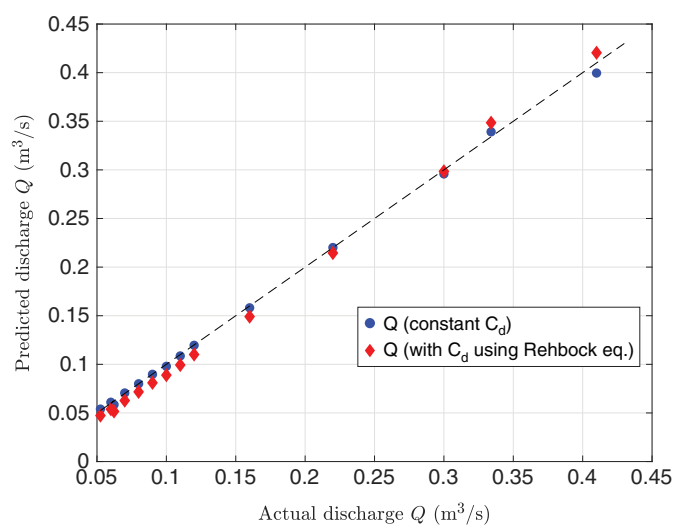

Fig. 8. (Color) Comparison of predicted discharges using Eq. (1) with $C_d = 0.75$ and C_{dR} computed using Eq. (2) versus actual discharge. The dashed line shows the one-to-one correlation.

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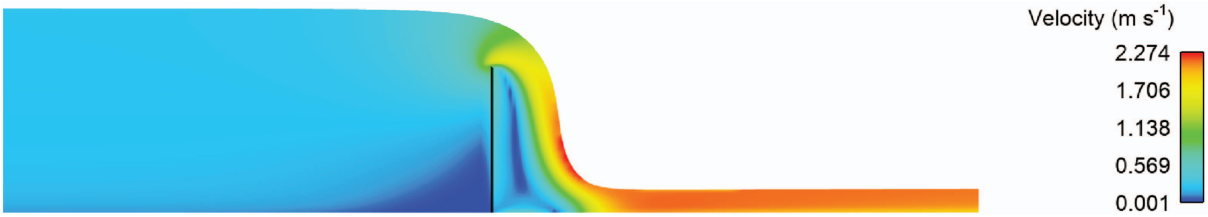

Fig. 9. (Color) Contours of magnitude of simulated resultant velocity for $h/P = 0.38$ depicting a clinging flow conditions due to the low relatively low head over the weir.

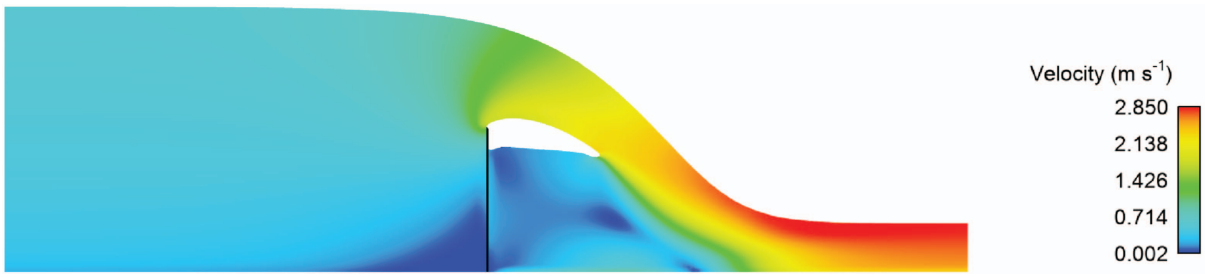

Fig. 10. (Color) Contours of magnitude of simulated resultant velocity for $h/P = 0.81$ depicting an ideal operating flow condition over the weir.

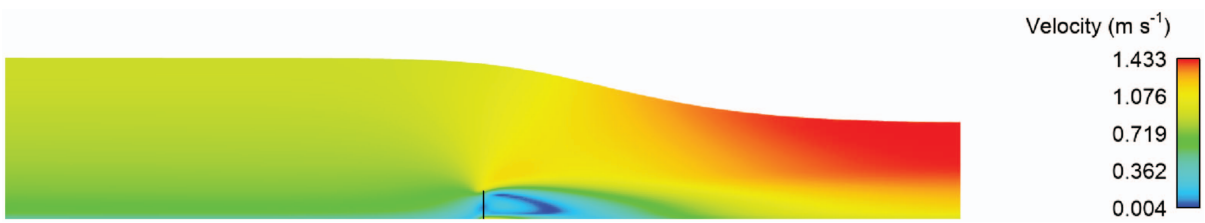

Fig. 11. (Color) Contours of magnitude of simulated resultant velocity for $h/P = 4.55$ depicting a weir-inundated flow condition.

Conclusions

An analysis of sharp-crested weirs using highly resolved CFD simulations revealed the ideal operating range for sharp-crested weirs in open-channel flow. Velocity and pressure profiles as well as discharge coefficients were analyzed to identify three separate flow regimes based on h/P values: a high acceleration regime, an ideal operating regime, and a weir-inundated regime.

For $h/P \leq 0.6$, flow close to the weir crest will experience high acceleration, resulting in negative pressure regions and a tendency towards clinging flow. Furthermore, the discharge coefficient experiences a negative linear trend as h/P increases to 0.60. In this regime, actual discharge coefficient values are markedly higher (about 10% to 40%) than the commonly used Rehbock values, leading to potentially serious mismanagement of water allocation and accounting in the field.

Between h/P of 0.6 and 2.0, assumptions built into the derivation of the discharge equation appear to hold, and the discharge coefficient becomes relatively constant. This operating regime enhances the accuracy of the weir for flow measurement and regulation.

Beyond this, at $h/P > 2.0$, flow begins to drown out the weir's influence. Velocity profiles begin to return to those representative of developed boundary layer flow, and pressure profiles trend toward hydrostatic conditions. The discharge coefficients fluctuate before increasing rapidly. It was also found that the Rehbock equation is a poor approximation for the discharge coefficient in this weir operating range.

Data Availability Statement

Data used during this study are available from the corresponding author upon reasonable request.

Acknowledgments

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Notation

The following symbols are used in this paper:

b = length of weir in spanwise direction (normal to flow);
C_d = discharge coefficient from classical rating equation [Eq. (1)];
C_{dT} = discharge coefficient obtained using the total head [Eq. (3)];
C_{dR} = estimated discharge coefficient from Rehbock equation [Eq. (2)];
g = gravitational acceleration constant;

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H = total mechanical energy head above weir crest;
h = elevation head above weir crest;
P = weir height;
p = pressure;
p_h = hydrostatic pressure;
Q = flow rate;
Re = Reynolds number;
U = reference velocity;
u = x-direction (streamwise) velocity;
W_c = crest width of weir in direction of flow;
We = Weber number;
X = horizontal distance;
Y = vertical distance;
y = depth of flow above weir crest;
$\bar{y}$ = depth of flow at Section 0-0;
$y+$ = nondimensional wall unit; and
γ = specific weight of water.

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